

Science + Tech -> Gate Mouth (PRI)

How science and tech frontiers map onto Gate as a clearance mouth - contribute under coordinators, do not own their monopolies.

Source: *gate/SCIENCE.md*

Motive

Contribute massively to Tier-S continuity with global coordinators. Collaborate as the clearance mouth. Do not own C2, nukes, grid, or link monopolies.

Formula

Inhibit unjustified irreversible acts (PRI) - fail-closed - as a driver-node mouth on the act graph, not a hub dashboard.

Tech thesis

Frontiers multiply what systems can do. Gate owns the mouth between can and may - contribute under each stack; do not own their guns.

First weld (named)

Withdraw / payout - clear before wire.

- Prove DENY on one irreversible money-leave path: POST /v1/payouts/{id}/release
- Checkout: /operator -> withdraw
- Next after prove: grid shed/reconnect (not first). Nuclear C2 stays state-only.

Science -> Gate law

PRI (pre-irreversibility inhibition)

execute iff LIVE AND justified; else DENY; no bypass; CHARGE-only re-open

Fail-closed

uncertainty => DENY; soft-yes under panic is the failure mode

Thermo intervention (no perfect do(.))

measure -> control -> stranger-verifiable receipt; silent free yes illegal

Driver nodes (Liu-Slotine-Barabasi)

weld irreversible edges; hubs are not drivers

Quorum / unfireable mouth

LIVE needs policy quorum + CHARGE + Gate; evidence outside actor trust boundary

Tech frontiers -> mouth

AI agent control / MCP firewalls

LIVE/DENY on irreversible agent writes; soft prompt is not DENY. Own? No - under operator tool graphs.

Grid-forming inverters + smart breakers

PRI on shed/reconnect edges after payout prove. Own? No - under utilities.

Military LEO mesh / SDN class

Optional fail-closed on irreversible link/priority/release. Own? No - under programs.

TEE + MPC + HSM (PQ custody)

Custody proves keys; Gate proves LIVE. Own? No - beside vaults.

Planetary-defense kinetic GNC

Conceptual PRI on unjustified commit. Own? No - state owns the shot.

Tier S - contribute ladder

- 1. Commercial driver weld (payout clear) - prove DENY
- 2. Critical infra mouth (grid shed / wholesale finality)
- 3. Named in coordinator COOP (clears via Gate) + cleared custodians
- 4. Defense/comms release mouth under programs - force_production_weld false until real

Nuclear football: state only. We may someday be a tiny component under them - never the sovereign.

Distribution

- Sell PRI to coordinators, not consumers
- Land: /operator (one door)
- First weld = payout clear; grid is next
- Find driver edges, not vanity hubs
- Price authority dissipation (weld + bps)
- Contracts/copy: collaborate; mouth only; favor to civilization
- Ads floor: /privacy + /terms; their_production false; pixels off by default

Machine-readable surfaces

GET /.well-known/science-pri.json | GET /science | GET /privacy | GET /terms | GET /.well-known/legal.json